Stationary Wavelet Convolutional Network With Generative Feature Learning for Hyperspectral Unmixing

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***Abstract*— Hyperspectral unmixing (HU) can obtain subpixel-level ground object information, which is crucial for the fine advancement of imaging spectrum processing technology. Deep learning (DL) has been widely used in HU recently because of its ability to deeply mine complex relevant features in data. Existing DL unmixing methods usually operate only in the original spatial–spectral feature domain. However, due to noise, spectral variation, and other factors, it is difficult to fully mine effective features and easy to interfere with by only relying on the original domain. To get over these obstacles, we propose an innovative stationary wavelet convolutional network (SWC-Net) for HU. Stationary wavelet transform (SWT) is introduced in SWC-Net to extend the original feature domain to feature domains with different frequencies, which promotes the multiview extraction of information. What is more, a new generative self-supervised feature learning strategy based on wavelet perspective (GSFL-W) is proposed for SWC-Net. More robust features can be obtained by GSFL-W by introducing noisy perturbations into high-frequency inputs and forcing the network to generate the original inputs. The proposed SWC-Net surpasses the advanced approaches by sufficient experiments on one simulated and three real hyperspectral datasets. The code is publicly available at https://github.com/UPCGIT/SWC-Net.**

***Index Terms*— Deep learning (DL), feature domain, generative self-supervised, hyperspectral unmixing (HU), stationary wavelet.**

I. Introduction

THE hyperspectral image (HSI) has the advantages of high spectral resolution and synchronous acquisition of spatial–spectral information, which is one of the important supporting data in earth observation applications. With the progress of imaging spectrometer technology, the HSI has found extensive use in diverse fields like environment monitoring [1], [2], tensor analysis [3], image classification [4], [5], [6], and object detection [7], [8]. However, since the spectral resolution and spatial resolution of sensors restrict each other, mixed pixels are ubiquitous in HSIs. Mixed pixels mean that multiple types of ground objects are contained in a pixel, which limits the role of HSIs in extracting fine ground object information. Hyperspectral unmixing (HU) is an effective method to deal with mixed pixels. From the perspective of the subpixel, HU aims to obtain the pure spectral signal (i.e., endmember) within the pixel and analyze the ratio of each category of endmembers within the pixel (i.e., abundance) [9].

To study the HU algorithm, the spectral mixing mechanism must first be expressed in mathematical form. The spectral mixing model can be divided into linear mixing model (LMM) and nonlinear mixing model (NLMM) [10]. LMM assumes that the mixture spectra are equal to the linearly weighted sum of the endmember spectra, where the weights are the corresponding abundances. The NLMM considers complex natural phenomena such as the propagation of light and theoretically will be closer to the real situation. However, it usually requires more parameters and prior knowledge, which are often difficult to obtain. Therefore, most of the existing unmixing methods are still based on the LMM due to its simple structure and strong versatility [11].

Traditional unmixing methods contain two independent processes: endmember extraction and abundance estimation. In this case, estimated abundance depends on the accuracy of endmember extraction, which is prone to error accumulation. To avoid this, statistical methods for estimating endmembers and abundance simultaneously have evolved considerably. Nonnegative matrix factorization (NMF) is one of the most representative methods, which is characterized by adding appropriate sparse [12], [13] or spatial [14], [15] constraints to matrix factorization. Unfortunately, too many constraints will increase the computational complexity, and the choice of constraints also requires a large amount of prior information.

In recent years, deep learning (DL) has brought many new ideas to HU with its flexible deep nonlinear architecture and powerful feature learning capabilities [16]. Well-designed autoencoder (AE) is the mainstream architecture to solve HU problems [17]. The AE includes an encoder for dimensionality reduction and a decoder for reconstructing the input. For HU, the hidden feature obtained by the encoder is the abundance,

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and the weight used by the decoder is the endmember [18]. AE-based HU frameworks can be mainly divided into two types: fully connected networks [19], [20], [21], [22], [23] and convolutional networks [24], [25], [26], [27], [28], [29]. Fully connected networks perform feature extraction in the spectral dimension. For example, Palsson et al. [21] applied a deep fully connected AE and innovative objective functions to HU, demonstrating the potential of DL in HU tasks. Jin et al. [23] introduced adversarial training into the AE and proposed an adversarial AE unmixing network for generating abundance close to ground-truth (GT) values. Unfortunately, such methods make it difficult to use spatial information, which is very important in image interpretation. Therefore, researchers are turning more attention to convolutional methods operating in spatial–spectral dimensions. The convolutional network method can fully extract spatial information through the sliding of spatial filtering, which breaks through the limitation that traditional methods cannot directly use spatial information. For instance, Gao et al. [26] combined cyclic consistent training with convolutional AE (CAE) to take full advantage of high-level semantic features in space. Yu et al. [27] proposed a scale fusion framework to mine information from multiple spatial scales and fuse it stage by stage. Yang et al. [28] adopted a U-shaped Transformer structure to capture global information and multiscale spatial information. Tao et al. [29] changed the architecture of considering decoder weights as endmembers and proposed a novel method to obtain endmembers using the multiplication of feature maps and abundance matrices.

From extracting spectral features to spatial–spectral features, more diversified features are proved to be an important means to improve DL unmixing. Therefore, how to mine more features is the problem faced by HU. Recently, many methods have emerged to further extract features using different approaches. For example, multiscale [27] and multihead self-attention [25], [28], but still only operate in the original feature domain, that is, edge and homogeneous regions with different frequencies are in the same image. The multimodal approach introduces a laser radar (LiDAR) perspective [30], but the HSI is still a single feature domain and it is difficult to obtain LiDAR data synchronously. Due to noise, spectral variation, and other factors, the mining difficulty and mixing mechanism of linear features with different frequencies are different, which may affect and interfere with each other. The ability to rely only on the original spatial–spectral domain with a single perspective to obtain accurate and distinguishable features is limited. Therefore, it is necessary to expand the diversified feature representation from the perspective of multidomain feature extraction to enhance the diversity of network learning.

Wavelet transform (WT) can transform the image from the original domain to the wavelet domain [31], which has been widely applied in computer vision [32], [33]. Recently, the WT has been tried to be applied to endmember bundle extraction [33], Bayesian unmixing [34], and NMF-based unmixing methods [35]. These methods transfer the unmixing task from the original spectral domain to the wavelet domain. In addition, Yan et al. [37] and Zhan et al. [38] tried to combine WT with a fully connected AE for unmixing. Most of the above methods use 1-D discrete WT (DWT) [39]. However, unmixing methods in the wavelet domain have just started and are not mature enough. Existing wavelet unmixing methods only consider the spectral-domain transformation, which cannot preserve the spatial structure of images. Moreover, DWT is easy to cause image distortion and information loss [40], [41]. Therefore, we propose a band sequential processing stationary WT (BSPSWT). BSPSWT improves the detail loss caused by DWT and can expand the original feature domain into one low-frequency domain and three high-frequency domains in different directions while retaining the 3-D spatial–spectral structure. On this basis, to deeply extract spatial–spectral information in transform feature domains of different frequencies for HU, we propose an innovative stationary wavelet convolutional network (SWC-Net). A parallel multidomain feature encoding module (MDFEM) is designed as the encoder backbone. MDFEM contains one low-frequency CAE branch and three high-frequency CAE branches. Information on different frequency domains can be captured by MDFEM, which helps to obtain richer and more differentiated features from different perspectives. The low-frequency branch can capture generalized features to grasp the whole, while the high-frequency branch can mine more detailed information to deeply understand the complex structure.

In addition, how to extract more effective features is also a problem to be solved. The effective feature in HU is the correct linear mapping between endmember and abundance. However, due to the nonconvexity of the HU task [42], the simple pursuit of reconstruction error minimization may lead to wrong abundances or endmembers, which is not sufficient to ensure accurate and effective features [26]. Simple identity mapping can be at risk of overfitting, especially if there is abnormal interference. Such overfitting often leads to the model incorrectly fitting abnormal interference as an effective feature to minimize the reconstruction error, thus reducing the extraction accuracy of the feature [43]. To overcome this challenge, a novel generative self-supervised feature learning strategy based on wavelet perspective (GSFL-W) is proposed. The motivation is that if the model can generate a raw signal from damaged data, it means that the model has learned the key features that represent the original signal. Specifically, abnormal interference such as noise or spectral variation belongs to high-frequency signals, which usually exist in high-frequency features after WT. The proposed GSFL-W strategy adds noise to high-frequency feature inputs and makes the network learn how to generate from the noisy image to the original image. In general, the proportion and contribution of abnormal interference in the original image is not large. Therefore, when the training difficulty increases, the high-frequency branch will first learn more effective and stable features for restoring the original image, thus reducing the influence of the original interference. In the wavelet framework of the proposed method, the more targeted GSFL-W strategy is applied to high-frequency features independently, and this flexibility makes the proposed network more adaptable to different scenarios.

The proposed SWC-Net mainly includes three parts: 1) BSPSWT is the proposed feature-domain extension method that can preserve the data structure; 2) MDFEM is designed for mining and fusing multiview information obtained from

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BSPSWT; and 3) GSFL-W achieves more effective feature extraction with an innovative high-frequency noise perturbation strategy. Based on the above, the contribution points are summarized in the following three aspects.

1) Based on the proposed BSPSWT to extend the diversified feature representation of HSIs, we develop a well-performing SWC-Net for HU. The MDFEM encoder is designed in the network to collaborate information from multiple feature spaces of spatial domain, spectral domain, and transform domain.
2) To solve the problem that identity mapping easily causes overfitting and takes interference as effective feature fitting, a generative self-supervised feature learning strategy based on wavelet is proposed. The proposed GSFL-W strategy considers that abnormal interference mainly exists in the high-frequency features after WT, forcing the network to extract more effective features to generate the original image from the noisy image of high-frequency branches.
3) The proposed SWC-Net fully explores the application of WT in multidomain feature representation for HU. Sufficient comparative experiments show the superiority of the proposed SWC-Net. The design rationality and necessity of the GSFL-W strategy are also demonstrated in the ablation experiment.

The subsequent structure is outlined hereunder. Section II delves into the pertinent prior work. Section III presents the proposed SWC-Net method. Section IV reports the experimental outcomes on simulated and real datasets. Finally, the overarching conclusion is summarized in Section V.

II. Related Works

Given that the proposed method relies on LMM, AE, and WT, a concise overview of pertinent studies will be provided in Section II.

A. AE-Based Method for LMM Unmixing

LMM presupposes that the spectral of each pixel can be viewed as a linear blend of endmember spectra, which is mathematically expressed as

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N} \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{L\times h\times w}$ represents the HSI with L bands, h lines, and w columns. $\mathbf{E} \in \mathbb{R}^{L\times p}$ represents the endmember matrix with p classes. $\mathbf{A} \in \mathbb{R}^{p\times h\times w}$ represents the abundance matrix of $\mathbf{Y}$. $\mathbf{N} \in \mathbb{R}^{L\times h\times w}$ is the noise matrix. LMM includes two commonly used constraints, namely, abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC). The mathematical expression is as follows:

$$a_j \geq 0 \tag{2}$$

$$\sum_{i=1}^{p} a_{ij} = 1 \tag{3}$$

where a_j is the abundance vector of the jth pixel, and i stands for the endmember category. Under the LMM framework, the AE-based unmixing process can be expressed by

$$\hat{\mathbf{Y}} = \sigma_D(w_D(\text{constraint}(\sigma_E(w_E\mathbf{Y})))) \tag{4}$$

where w_E and w_D are the weights of the encoder and the decoder, respectively. σ_E and σ_D are the activation functions of the encoder and the decoder, respectively. Constraint represents the ANC and ASC that need to be implemented. The encoder can have a flexible multilayer structure, which is equivalent to the feature extraction part. To conform to LMM, the decoder structure is relatively fixed. There is usually only one layer and σ_D is a linear function. The training goal of the AE for HU is to make the output $\hat{\mathbf{Y}}$ equal to the input $\mathbf{Y}$ as much as possible. In this design, the encoder output constraint$(\sigma_E(w_E\mathbf{Y}))$ can be regarded as abundance and decoder weight w_D as endmember.

B. Wavelet Transform

WT is originally used for signal time–frequency analysis, and its function is to decompose complex signals into a combination of simple basis functions with different coefficients. In a 2-D image, WT can decompose the image into low-frequency and high-frequency parts. Low frequency means that the pixel value changes slowly, which means that the area is homogeneous. High frequency refers to the large difference in pixel values between adjacent areas, which means that the area is an edge. For a single-band 2-D image $f(x, y)$, its WT expression is

$$f(x,y) = \sum_{j=1}^{J}\sum_{g,q,r}\left\langle f(x,y), \psi_{j,g,q,r}(x,y)\right\rangle \psi_{j,g,q,r}(x,y) + \left\langle f(x,y), \rho_j(x,y)\right\rangle \rho_j(x,y) \tag{5}$$

where $\psi_{j,g,q,r}(x, y)$ represents the wavelet basis function and J is the number of decomposition layers. $\rho_j(x, y)$ is the scale function. The scale function is used to describe the overall low-frequency variation of the signal over time or space. The wavelet basis function can be regarded as the result of translation and scaling of the scale function. Through different scale and shift operations, wavelet combinations with different coefficients can be constructed to adapt to the high-frequency changes of the signal. There are different types of wavelet basis functions, and their effects on HU will be evaluated in ablation experiments. (g, q, r) is the amount of translation in the horizontal (H), vertical (V), and diagonal (D) directions. $\langle f(x, y), \psi_{j,g,q,r}(x, y)\rangle$ represents the scalar product of the image $f(x, y)$ with the wavelet function of scale 2^j in H-, V-, and D-directions, representing the high-frequency coefficient of the image in different directions. $\langle f(x, y), \rho_j(x, y)\rangle$ represents the scalar product of the image with the scale function, representing the low-frequency coefficient of the image. So, $\langle f(x, y), \psi_{j,g,q,r}(x, y)\rangle\psi_{j,g,q,r}(x, y)$ is the high-frequency feature in the three directions obtained by WT, and $\langle f(x, y), \rho_j(x, y)\rangle\rho_j(x, y)$ is the low-frequency feature. Two-dimensional WT essentially maps the original image to different frequency domains through a combination of basis functions and corresponding coefficients. It should be noted that multilayer decomposition (i.e., $j > 1$) is to further

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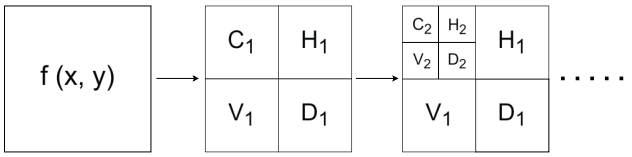


Fig. 1. Process of DWT. C_j represents the low-frequency feature. H_j, V_j, and D_j represent the high-frequency features in different directions.

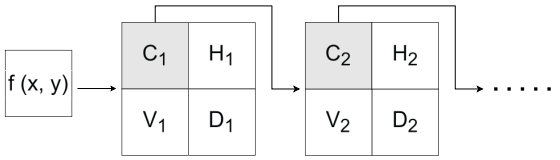


Fig. 2. Process of SWT. C_j represents the low-frequency feature. H_j, V_j, and D_j represent the high-frequency features in different directions.

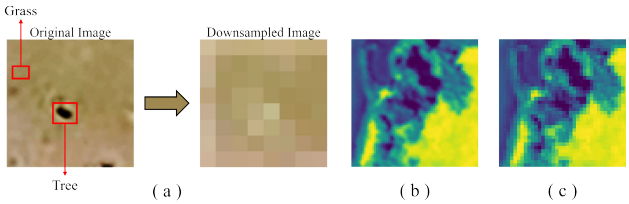


Fig. 3. (a) Downsampling causes small target loss. (b) Coefficients obtained by SWT. (c) Coefficients obtained by DWT.

decompose the low-frequency feature obtained in the previous layer.

The difference between DWT and stationary WT (SWT) [44], [45] is the operation on the initial coefficient size. For the jth layer, since the scale of the wavelet function and the scale function is 2^j, that is, the sampling step is 2^j, the initial coefficient size obtained is $1/2^j$ of the original image. The scale is downsampled in DWT, and the final wavelet coefficient and corresponding feature size is $1/4^j$ of the original image. The purpose is to reduce the amount of data while maintaining the main characteristics of the signal, which is applicable in signal processing. Fig. 1 illustrates the DWT flow. However, in natural image processing, downsampling may lead to the loss of small targets. DWT also distorts the edge part of the image, producing information loss. Therefore, SWT is proposed to solve this problem. SWT upsamples the initial coefficients so that the size of each feature component is consistent with the original image. The SWT process is shown in Fig. 2. Fig. 3 shows downsampling on small targets and the coefficients obtained by DWT and SWT, respectively. As can be seen from the figure, the coefficients obtained by DWT have obvious jagged distortion on the edges.

III. Proposed Method

The overall structure of the proposed method is represented in Fig. 4. To preserve the original spatial–spectral structure from the original domain to the wavelet domain, BSPSWT is designed on HSIs to obtain more refined features at first, and the original single-frequency spatial–spectral feature is expanded into one low-frequency feature and three high-frequency features. Next, to extract more effective information from high-frequency features that contain lots of complex details, the GSFL-W strategy of mining good representations from perturbed high-frequency features is proposed. Finally, different transform domain features use their dedicated CAE branches to mine and fuse spatial–spectral structures and the designed MDFEM can more fully characterize the input data.

A. BSPSWT for HSIs

A simple and low computational band sequential processing flow is designed and performed to apply the 2-D SWT to 3-D HSIs to obtain features in different frequency domains while preserving the original structure. Fig. 5 presents the flowchart of the process. As can be seen from the figure, BSPSWT essentially performs SWT for each band individually and combines them sequentially. Before being performed SWT, $\mathbf{Y} \in \mathbb{R}^{L\times h\times w}$ needs to be padded to make sure that lines and columns can be divided by 2. This is essential because the step size of the WT sampling is a multiple of 2. Padding is filling the surrounding pixels with edge values. After executing SWT, the fill pixel is removed. Therefore, this filling process does not affect subsequent network training.

Increasing the number of decomposition layers will greatly increase the complexity of the network. Therefore, the proposed method adopts one-layer SWT decomposition, that is, four characteristic components are obtained. BSPSWT processes the 2-D images of all bands band by band to obtain four frequency-domain features for each band. The resulting same frequency-domain features are then connected separately. Finally, the characteristics of HSIs in four frequency domains are generated. The process can be represented by the formula as follows:

$$\begin{aligned}\text{Connect}_{i=1}^{L}(\text{SWT}(y_i)) &= \text{Connect}_{i=1}^{L}(\text{LF}_i, \text{HF}_i, \text{VF}_i, \text{DF}_i)\\ &= (\text{LF}, \text{HF}, \text{VF}, \text{DF})\end{aligned} \tag{6}$$

where LF represents the low-frequency features. HF, VF, and DF represent the high-frequency features in different directions. y_i represents the HSI image of the ith band. Connect is the connection operation band by band. It is not difficult to see from Fig. 5 that the low-frequency features retain the overall information of the original image. High-frequency features contain detailed information in different directions. In this way, features preserving the spatial–spectral structure in different frequency domains are obtained. Richer features can help the network fully understand the internal structure and content of the image.

B. GSFL-W Strategy

Fig. 6 takes Samson's data as an example to show the different spectral curves of the same ground objects in a certain region. It is not difficult to find that even the same object will have small differences. These differences may be caused by noise or spectral variation. Regular AE networks will try their best to fit all the details, resulting in overfitting. The abnormal information in the difference will affect the accuracy of feature extraction, as shown in Fig. 7. Generative self-supervised learning [46], [47] is to artificially add noise perturbations to input data and achieve feature learning by forcing the model to generate original data from noisy data. The increase in training

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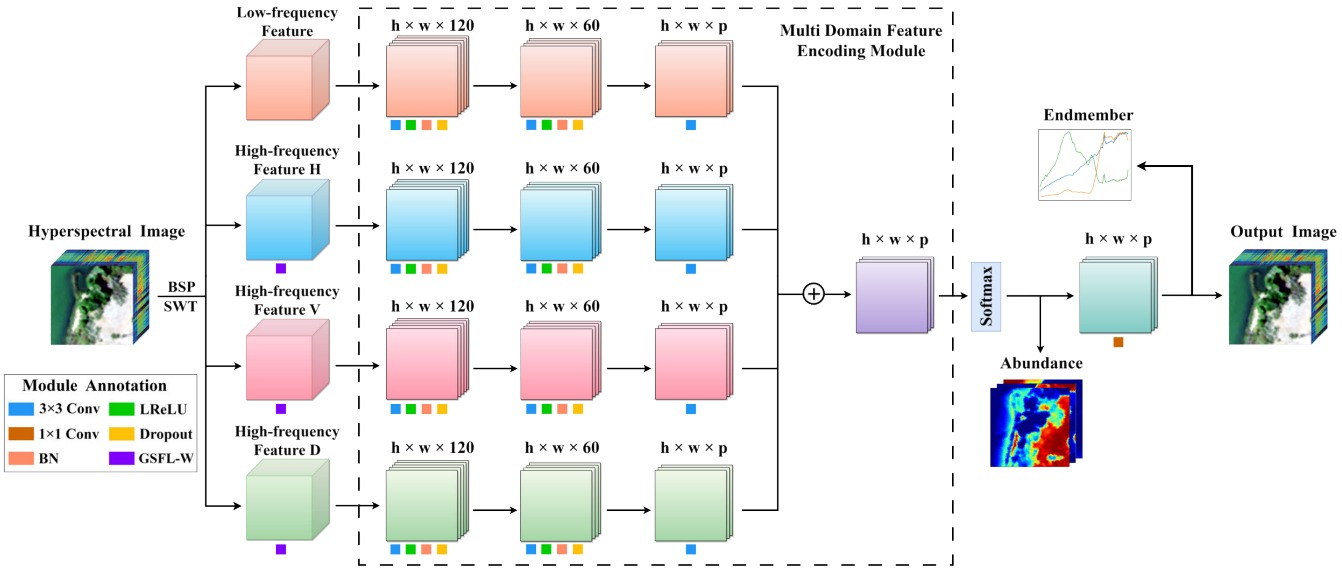


Fig. 4. Overview of the SWC-Net. The features of different frequency domains are input into their respective CAE branches. The GSFL-W strategy is executed in the high-frequency branches.

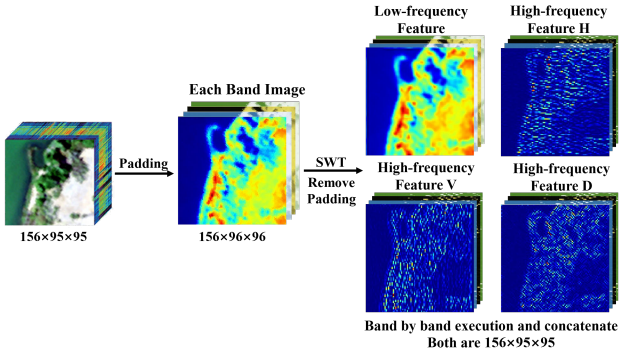


Fig. 5. Flowchart of BSPSWT for HSIs.

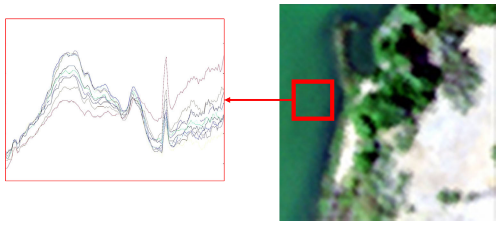


Fig. 6. Different spectral curves of the same feature, taking Samson as an example.

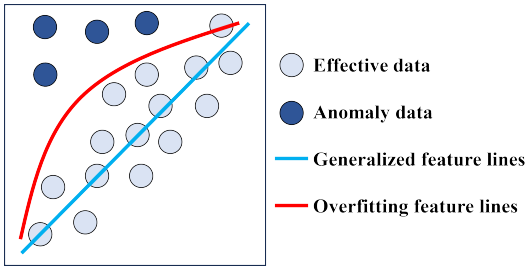


Fig. 7. Overfitting will cause feature bias due to the interference of abnormal data.

difficulty can avoid overfitting, which is essentially a form of data augmentation [48]. Fig. 8 illustrates the difference between regular AE training and generative self-supervised

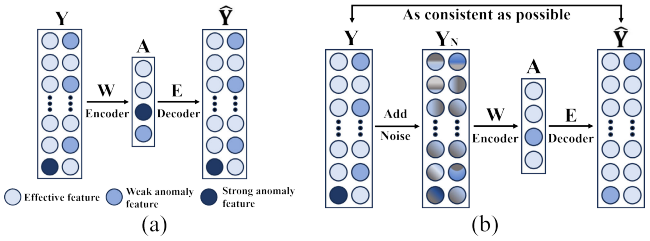


Fig. 8. (a) Regular AE. (b) Generative self-supervised learning.

training. In the figure, different degrees of blue represent the effective and abnormal features of a ground object. Under normal identity mapping, the network will simultaneously learn the effective and abnormal features. In generative self-supervised learning, where the difficulty of mapping increases, and the contribution of abnormal features to the original data is small. In this case, the network is forced to prioritize learning how to generate effective features from perturbing features, rather than trying to remember the specific details in the data. What is more, the network also learns a variety of mappings to generate effective features from the perturbation feature, including the original perturbation to a certain extent. Therefore, the possibility of the network learning a small number of abnormal features will become low, and this part will eventually be mapped by the effective mapping relationship that the network has learned to generate accurate features. Although the increased training difficulty means that the output is not necessarily able to be the same as the input at this time, it can obtain more robust hidden layer characteristics, that is, the required abundance and endmember.

The high-frequency features after WT contain a more complex feature mapping relationship, in addition, it often contains noise. Therefore, high-frequency features are more likely to contain abnormal interference data, thereby impeding the network's ability to accurately discern the correct mapping between the endmember and the abundance. On the contrary,

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low-frequency feature has less interference, and adding noise to them may introduce too much interference. The proposed GSFL-W strategy takes into account the different characteristics of low-frequency and high-frequency features and only adds noise disturbance to the input of the high-frequency branch. In this article, Gaussian noise is selected because it conforms to the normal distribution and can avoid overfitting specific details. The shape of the noise is consistent with the high-frequency features. The Gaussian noise is summed up with the three high-frequency features before being inputted into the network. The process can be represented as follows:

$$\text{SWCNet}(\text{LF}, \text{HF}+\text{GN}, \text{VF}+\text{GN}, \text{DF}+\text{GN}) = \hat{\mathbf{Y}} \quad (7)$$

where GN represents the Gaussian noise. The goal is to make the output $\hat{\mathbf{Y}}$ consistent with the original HSI. Unlike the existing methods, the GSFL-W strategy aims to extract the main effective features from the high-frequency components and prevent overfitting to the abnormal details.

C. Network Architecture

The network includes an MDFEM encoder backbone which consists of four branches and feature fusion, as well as a linear decoder. Each encoder branch receives a frequency-domain feature as input. The four branches of the encoder have the same three-layer structure. The first encoder layer incorporates a 2-D convolutional operation, reducing the spectral channel dimension to 120 and adopting a 3×3 sliding window size. Leaky ReLU is used as a nonlinear activation function, ensuring that inputs below 0 are not neglected. Batch normalization (BN) is then implemented to expedite the training and mitigate potential exponential changes in weights. Finally, dropout with a ratio of 0.1 is used to avoid overfitting. The second encoder layer is consistent with the initial layer and continues to reduce the dimension to 60. The final encoder layer reduces the channel dimension to p and no more activation function, BN, and dropout. The four branches end up with four hidden layer features, which are fused by the adding method. The features obtained from MDFEM are then fed into the softmax function. Softmax has the capability of multiclass classification, which can transform the input into a probability (Pr) distribution. Each probability value is between 0 and 1, and its sum equals 1. Assumes that $\vec{z} = [z_1, z_2, \ldots, z_K]$ is the input vector, its softmax expression is

$$\text{softmax}\left(\vec{z}\right)_i = \frac{e^{z_i}}{\sum_{i=1}^{K} e^{z_i}} = [\Pr(z_1), \Pr(z_2), \ldots, \Pr(z_K)]$$
$$0 < \Pr(z_i) < 1, \quad \Pr(z_1) + \Pr(z_2) + \cdots + \Pr(z_k) = 1. \quad (8)$$

This property conforms to ANC and ASC and is, therefore, used to obtain abundance. Finally, the hidden layer features need to be reconstructed into the input through the decoder weights (i.e., endmember matrix), using a 2-D convolution with 1×1 filter and L feature mappings. No nonlinear activation function is used to satisfy the LMM. Leveraging the high precision of spectral angle distance (SAD) in capturing HSI variations within high-dimensional structures, it is adopted as the loss function. The formula is

$$\text{Loss} = \arccos\left(\frac{\langle \mathbf{Y}, \hat{\mathbf{Y}} \rangle}{\|\mathbf{Y}\|_2 \left\|\hat{\mathbf{Y}}\right\|_2}\right). \quad (9)$$

IV. Experiments

All experimental results are obtained after ten repeated runs. The outcomes are presented as "average ± standard deviation." The average value represents the accuracy of the method, that is, whether it can produce correct unmixing results. The standard deviation represents the consistency of the method, that is, whether it can produce the same result in each run.

For comparative experiments, vertex component analysis (VCA) [49] and $L_{1/2}$-NMF [12] are chosen as representatives of traditional approaches. Six state-of-the-art DL methods are also selected: deep AE unmixing (DAEU) [21], adversarial AE network (AAE-Net) [23], cycle consistency unmixing network (CyCU-Net) [26], multistage network (MS-Net) [27], U-shaped Transformer network (UST-Net) [28], and abundance-guided spectral and spatial networks (A2SN) [29]. The first two are based on a fully connected network, and the remaining four including the proposed method are based on a convolutional network.

A. Datasets Introduction

One simulated and three real data are used in the experiment, as presented in Fig. 9. The details are elaborated below.

1) Simulated Data: To verify the noise resistance, we generate simulation data. Five endmembers' spectra for a simulated dataset with 166 bands are extracted in the USGS digital spectral library [50]. The abundance maps with the spatial size 130×130 are generated through the hyperspectral data retrieval and analysis (HYDRA) toolbox through the Gaussian field method.

2) Samson: The Samson dataset is a popular real dataset, which is often used to test the performance of HU methods. It is obtained by the Spectroscopic Aerial Mapper with Onboard Navigation (SAMSON) sensor [51]. It contains 95×95 pixels, featuring a spectrum of 156 bands, covering a wavelength range extending from 401 to 889 nm. There are three main materials: soil, tree, and water.

3) Houston: This famous real dataset is originated from the 2013 IEEE GRSS Data Fusion Contest and later used for HU evaluation [24]. It comprises a 105×105 pixel image with 144 bands and wavelength coverage ranging from 380 to 1050 nm. The data comprises four primary components: grass, runway, parking lot 1, and parking lot 2.

4) Moffett: This commonly used real hyperspectral dataset is obtained from the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and is a challenging unmixing dataset [52]. It contains 50×50 pixels, featuring a spectrum of 184 bands. There are three main materials: water, soil, and vegetation.

The reference GT for endmembers of the real datasets is the pixel spectrum selected in the image that is close to the actual situation [53]. The reference GT for abundance maps is generated using fully constrained least-squares according to the endmember GT [54].

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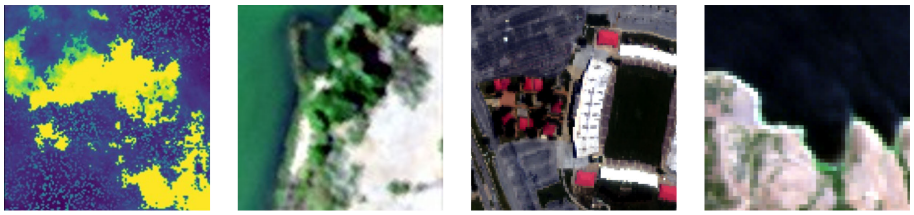


Fig. 9. Visualization of datasets: (a) simulated data, (b) Samson, (c) Houston, (d) Moffett.

TABLE I
NETWORK PARAMETER VALUE SETTING

	LR	Weight decay	S	G
Simulated data	0.004	5e-5	45	0.8
Samson	0.03	2e-4	35	0.7
Houston	0.005	1e-3	40	0.8
Moffett	0.02	5e-4	35	0.5

B. Experimental Setup

1) Implementation Details: The basic settings for training DL networks are optimizer, learning rate (LR), and epochs. In the experiment, the Adam incorporating weight decay is selected as the optimizer. LR adopts an exponential decay strategy, namely, it is scaled down by factor G after each S epoch. The number of epochs is set to 400 for Samson and 300 for others. The parameter values are set in Table I. It should be noted that DL HU is usually a blind HU task that does not require a partitioning training set and test set. The HSI is input into the network as a whole rather than as a patch. This is because blind HU is an unsupervised task, meaning that HU does not require artificial labels.

2) Assessment Indicators: For quantitative analysis, two classical assessment indicators are used to test the accuracy of methods: SAD and root mean square error (RMSE), and the mathematical expressions are

$$\mathrm{SAD}(\hat{e}_i, e_i) = \arccos\left(\frac{\hat{e}_i e_i}{\|\hat{e}_i\|_2 \|e_i\|_2}\right) \tag{10}$$

$$\mathrm{RMSE}(\hat{a}_j, a_j) = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\|\hat{a}_j - a_j\|_2^2} \tag{11}$$

where $\hat{e}_i$ and e_i are the resulting endmember and GT, respectively. $\hat{a}_j$ and a_j represent the resulting abundance and GT, respectively. Both SAD and RMSE are smaller, the better. N is the total number of pixels.

C. Noise Robustness Analysis With Simulated Data

To evaluate the noise robustness of the proposed SWC-Net, the simulated data is introduced with 10, 20, and 30 dB noise. Table II shows the corresponding results for different signal-to-noise ratio (SNR) levels. It can be seen from the table that the proposed method always maintains satisfactory performance under three SNR conditions and reaches the best average values in terms of endmember extraction and abundance inversion, which has good robustness. In addition, it can also be found that the convolutional methods have better performance as a whole. From a high SNR (SNR = 30 dB) to a medium SNR (SNR = 20 dB), the performance degradation of most methods is not drastic. After reducing to a low SNR (SNR = 10 dB), the performance of most methods has been significantly reduced, but the proposed SWC-Net still has the optimal results. This demonstrates the strong potential of the wavelet architecture to resist noise interference.

D. Experiment With Samson

Table III shows the quantitative evaluation outcomes. It is not difficult to find that the method considering spatial information obviously exceeds the traditional and fully connected network method. The proposed SWC-Net has the optimal quantitative result, which shows that the combination of multi-frequency domain information further improves the unmixing performance. For intuitive comparison, Fig. 10 presents the visual comparison of extracted endmembers via various methods and reference GT, and the extracted abundance obtained by all methods is depicted in Fig. 11. As seen from the figure, the results obtained by SWC-Net are visually very consistent with GT and can provide convincing information. The results of MS-Net, UST-Net, and A2SN are also satisfactory, indicating that the structure of Samson may be relatively simple.

E. Experiment With Houston

Table IV shows the quantitative evaluation results of SAD and RMSE. The proposed SWC-Net still has the best quantitative evaluation results. The second-best SAD is obtained by the traditional method $L_{1/2}$-NMF, which indicates that the ground object distribution in Houston may have some sparsity. The RMSE result of SWC-Net is improved by 35.3% compared to the second-best result (i.e., MS-Net). This indicates that the abundance extraction of Houston has some difficulty. With the advantages of WT and generative self-supervised learning, the proposed method mines the effective features of more views and achieves better results in abundance and endmember extraction. Fig. 12 presents the visual comparison of extracted endmembers via various methods and reference GT, and the extracted abundance obtained by all methods is described in Fig. 13. As seen from the figure, the results obtained by the proposed method are visually similar to GT.

F. Experiment With Moffett

Table V shows the quantitative evaluation results. SWC-Net still maintains the best SAD and RMSE, indicating that

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TABLE II
SAD AND RMSE FOR THE SIMULATED DATA WITH DIFFERENT SNR. BOLD REPRESENTS THE OPTIMAL OUTCOMES

Method		VCA	$L_{1/2}$-NMF	DAEU	AAE-Net	CyCU-Net	MS-Net	UST-Net	A2SN	SWC-Net
SAD	SNR = 10 dB	0.114±0.028	0.120±0.021	0.250±0.018	0.178±0.019	0.105±0.023	0.225±0.014	0.227±0.040	0.089±0.011	**0.060±0.007**
	SNR = 20 dB	0.083±0.016	0.080±0.019	0.188±0.044	0.209±0.022	0.059±0.019	0.092±0.030	0.140±0.022	0.076±0.006	**0.036±0.004**
	SNR = 30 dB	0.082±0.007	0.067±0.009	0.156±0.043	0.225±0.025	0.056±0.013	0.069±0.008	0.105±0.017	0.072±0.008	**0.032±0.002**
RMSE	SNR = 10 dB	0.174±0.025	0.177±0.018	0.188±0.016	0.319±0.030	0.152±0.011	0.152±0.014	0.210±0.021	0.180±0.029	**0.135±0.010**
	SNR = 20 dB	0.137±0.023	0.133±0.013	0.200±0.032	0.334±0.030	0.109±0.020	0.111±0.016	0.204±0.018	0.132±0.015	**0.087±0.003**
	SNR = 30 dB	0.130±0.016	0.123±0.013	0.216±0.025	0.332±0.033	0.103±0.012	0.105±0.014	0.178±0.013	0.117±0.013	**0.082±0.002**

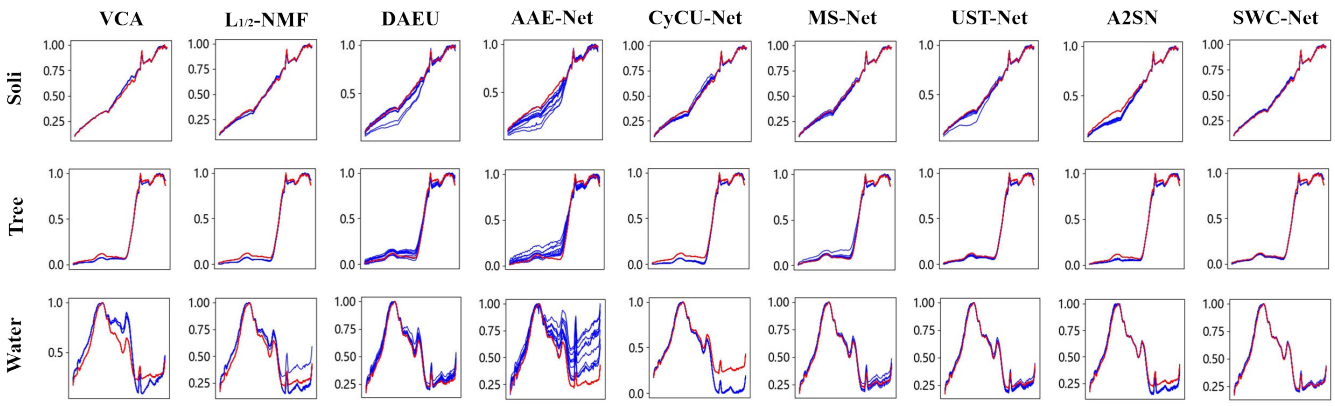


Fig. 10. Visualization of resulting endmember and GT on Samson. Blue line: extracted endmember. Red line: reference GT.

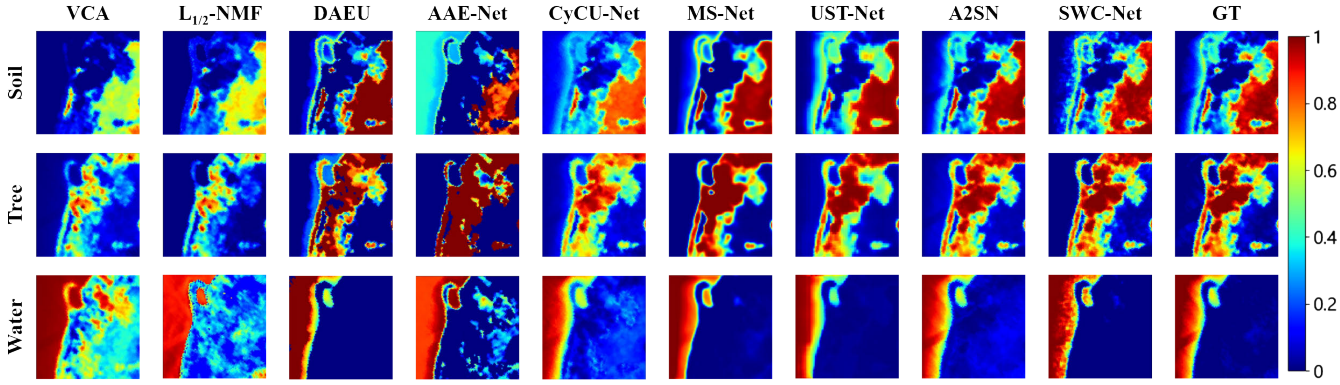


Fig. 11. Visualization of resulting abundance and GT on Samson.

TABLE III
SAD AND RMSE FOR THE SAMSON. BOLD REPRESENTS THE OPTIMAL OUTCOMES

Method		VCA	$L_{1/2}$-NMF	DAEU	AAE-Net	CyCU-Net	MS-Net	UST-Net	A2SN	SWC-Net
SAD	Soil	0.024±**0.001**	0.031±0.002	0.052±0.052	0.111±0.061	0.033±0.005	0.018±0.006	0.016±0.007	0.064±0.009	**0.015**±0.004
	Tree	0.041±0.002	0.052±0.002	0.067±0.021	0.100±0.051	0.073±0.003	0.045±0.024	0.034±0.002	0.050±**0.001**	**0.032**±0.002
	Water	0.154±0.005	0.094±0.004	0.045±0.014	0.237±0.100	0.260±0.005	0.032±0.009	0.036±0.005	0.077±0.004	**0.025±0.002**
	Average	0.073±0.002	0.059±0.002	0.055±0.022	0.149±0.043	0.122±**0.001**	0.032±0.011	0.029±0.002	0.064±0.003	**0.024**±0.002
RMSE	Soil	0.267±**0.001**	0.173±0.002	0.143±0.050	0.355±0.099	0.130±0.005	0.112±0.054	0.073±0.010	0.064±0.004	**0.061**±0.004
	Tree	0.251±0.002	0.176±0.002	0.157±0.067	0.332±0.124	0.108±0.006	0.102±0.051	0.069±0.013	0.042±**0.001**	**0.041**±0.002
	Water	0.425±**0.001**	0.527±0.005	0.140±0.094	0.191±0.121	0.133±0.006	0.053±0.019	**0.043**±0.010	0.060±0.003	0.047±0.005
	Average	0.314±**0.001**	0.292±0.002	0.147±0.036	0.293±0.069	0.124±0.004	0.089±0.040	0.062±0.009	0.055±0.003	**0.050**±0.002

it has good generalization in different real scenarios. The SAD result of SWC-Net is improved by 51.3% compared to the second-best result (i.e., UST-Net). There is little gap between the mean RMSE and the suboptimal method (i.e., VCA), but the mean SAD value is improved by 83.8%. This shows that it is difficult to extract the endmembers of

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TABLE IV
SAD AND RMSE FOR THE HOUSTON. BOLD REPRESENTS THE OPTIMAL OUTCOMES

Method		VCA	$L_{1/2}$-NMF	DAEU	AAE-Net	CyCU-Net	MS-Net	UST-Net	A2SN	SWC-Net
SAD	Grass	0.051±0.002	**0.037±0.001**	0.125±0.036	0.102±0.024	0.089±0.020	0.085±0.003	0.098±0.003	0.066±0.022	0.058±0.002
	Runway	0.137±0.063	0.167±0.022	0.086±0.014	0.196±0.044	0.207±0.028	0.080±0.036	**0.060**±0.005	0.263±0.030	0.099±**0.004**
	Parking lot 1	0.056±**0.001**	**0.053**±0.002	0.095±0.039	0.079±0.016	0.101±0.016	0.074±0.025	0.080±0.013	0.181±0.043	0.059±0.002
	Parking lot 2	0.117±0.080	**0.071**±0.014	0.221±0.094	0.367±0.198	0.197±0.017	0.129±0.038	0.168±0.008	0.490±0.314	0.083±**0.007**
	Average	0.090±0.019	0.082±0.006	0.132±0.027	0.186±0.053	0.148±0.009	0.092±0.013	0.102±0.004	0.250±0.096	**0.075±0.002**
RMSE	Grass	0.144±0.101	0.062±**0.002**	0.119±0.090	0.131±0.062	0.120±0.034	0.068±0.007	0.083±0.014	0.157±0.061	**0.045**±0.005
	Runway	0.086±0.048	0.059±0.004	0.107±0.022	0.184±0.041	0.266±0.059	0.054±0.003	0.070±0.022	0.223±0.037	**0.042±0.002**
	Parking lot 1	0.147±0.033	0.115±**0.004**	0.216±0.059	0.259±0.040	0.311±0.042	0.186±0.042	0.252±0.060	0.195±0.034	**0.111**±0.008
	Parking lot 2	0.194±0.021	0.256±**0.004**	0.159±0.057	0.303±0.041	0.323±0.034	0.156±0.035	0.216±0.068	0.216±0.051	**0.101**±0.008
	Average	0.143±0.043	0.123±**0.001**	0.150±0.036	0.219±0.032	0.255±0.017	0.116±0.017	0.155±0.026	0.198±0.021	**0.075**±0.003

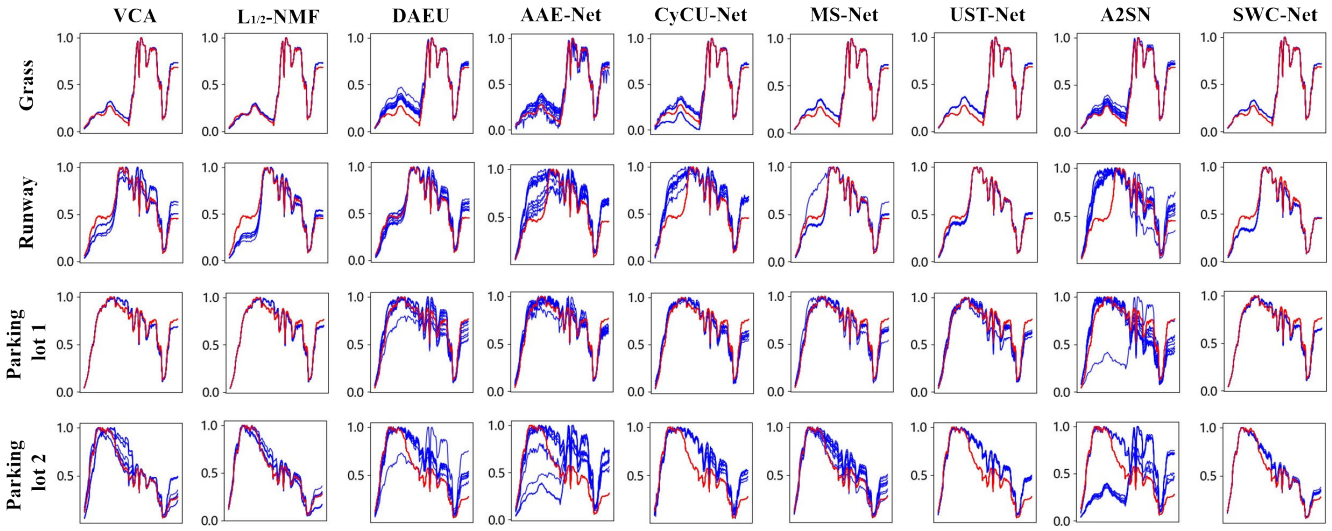


Fig. 12. Visualization of resulting endmember and GT on Houston. Blue line: extracted endmember. Red line: reference GT.

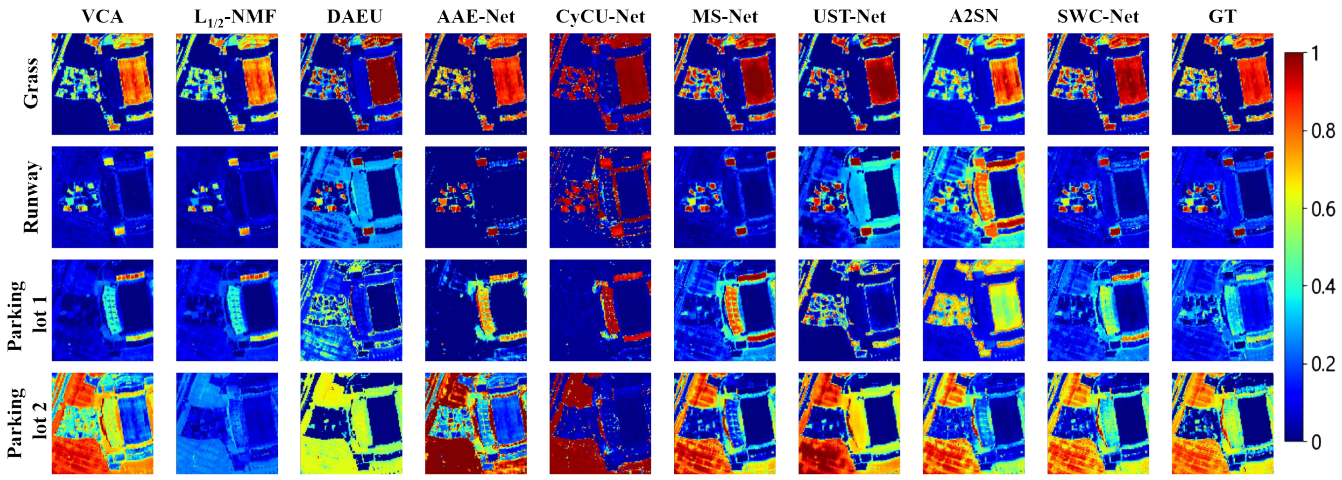


Fig. 13. Visualization comparison of resulting abundance and GT on Houston.

Moffett accurately. Fig. 14 presents the visual comparison of extracted endmembers via various methods and reference GT, and the extracted abundance obtained by all methods is described in Fig. 15. As seen from the figure, the proposed method can accurately extract endmembers in multiple runs.

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TABLE V
SAD AND RMSE FOR THE MOFFETT. BOLD REPRESENTS THE OPTIMAL OUTCOMES

Method		VCA	$L_{1/2}$-NMF	DAEU	AAE-Net	CyCU-Net	MS-Net	UST-Net	A2SN	SWC-Net
SAD	Water	0.189±0.024	0.379±0.071	0.054±0.032	0.071±0.078	0.033±0.025	0.020±**0.001**	**0.019**±0.002	0.024±0.003	0.023±0.006
	Soil	0.035±0.017	0.043±0.004	0.057±0.045	0.112±0.018	0.027±0.010	0.031±0.027	0.025±0.007	0.030±0.022	**0.012±0.001**
	Vegetation	0.108±0.013	0.136±**0.002**	0.200±0.138	0.569±0.296	0.142±0.045	0.060±0.034	0.068±0.009	0.148±0.185	**0.018**±0.003
	Average	0.111±0.009	0.186±0.023	0.104±0.065	0.251±0.120	0.068±0.020	0.038±0.020	0.037±0.004	0.067±0.068	**0.018±0.002**
RMSE	Water	0.075±0.005	0.445±0.012	0.173±0.125	0.089±0.019	0.080±0.006	0.085±0.011	0.086±0.009	0.140±0.114	**0.071±0.004**
	Soil	**0.047±0.002**	0.079±0.003	0.212±0.107	0.287±0.036	0.158±0.037	0.138±0.064	0.076±0.025	0.094±0.069	0.055±0.003
	Vegetation	0.088±0.004	**0.051**±0.004	0.265±0.134	0.266±0.045	0.165±0.045	0.121±0.058	0.075±0.027	0.138±0.126	0.054±**0.003**
	Average	0.070±**0.001**	0.192±0.003	0.217±0.111	0.214±0.028	0.134±0.029	0.114±0.041	0.079±0.017	0.124±0.103	**0.060**±0.002

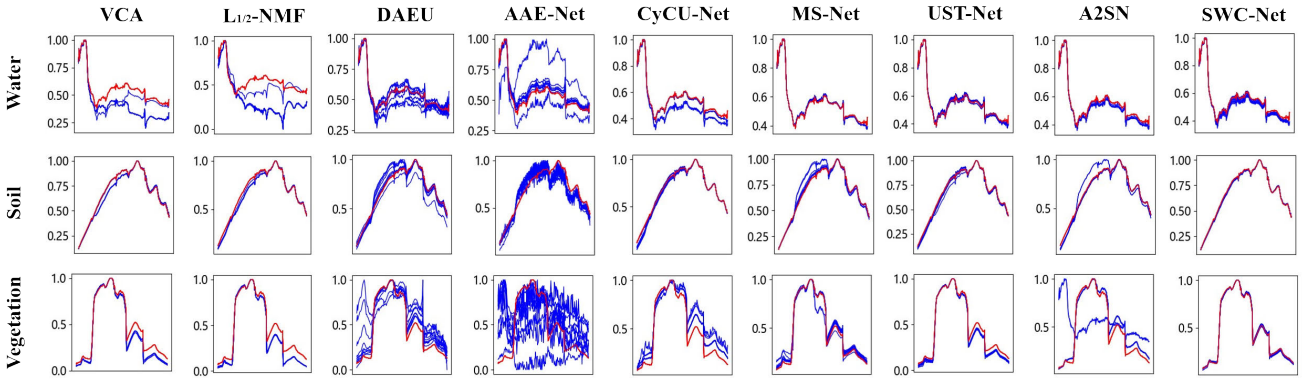


Fig. 14. Visualization of resulting endmember and GT on Moffett. Blue line: extracted endmember. Red line: reference GT.

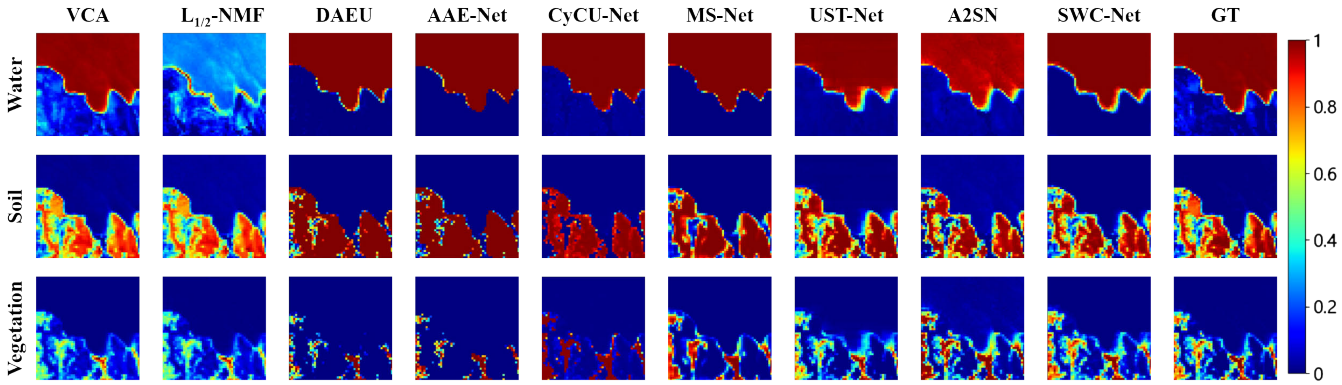


Fig. 15. Visualization of resulting abundance and GT on Moffett.

G. Ablation Experiment

Since the downsampling of DWT is easy to cause information loss, SWT is used in the proposed method. Generative self-supervised learning is to extract more robust features to resist interference, which usually exists in the high-frequency part, so GSFL-W adds noise in the high-frequency branch. To further verify the necessity of the above settings through data, relevant comparative experiments are performed. The results are shown in Table VI. Both SWT and DWT in the table use the network structure shown in Fig. 4 and the flow shown in Fig. 5, but the WT part takes a different approach and does not add GSFL-W. The purpose is to compare the difference between DWT and SWT. As you can see from the first two lines, SWT's results are significantly better than DWT's. More importantly, compared with the suboptimal results obtained in the comparison experiments on different datasets, SWC-Net still has the overall best performance when only SWT is used. This fully shows that it is effective and necessary to use WT to expand the diverse expression of features. GSFL-WL, GSFL-WH, and GSFL-WLH stand for adding noise in the low-frequency branch, high-frequency branch, and low and high-frequency branches, respectively. GSFL-WH is the GSFL-W strategy adopted in this article, so the fourth row in the table is the proposed SWC-Net. It is not difficult to find that the addition of noise in the low-frequency branch does not produce a significant change in performance compared to SWT. When the noise is added to the high-frequency branch, the unmixing accuracy is improved.

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TABLE VI
VALIDITY AND NECESSITY OF WT AND GSFL-W. BOLD REPRESENTS THE OPTIMAL OUTCOMES

	Simulated (20 dB)		Samson		Houston		Moffett	
	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
DWT	0.056±0.009	0.123±0.008	0.030±0.007	0.079±0.009	0.075±0.004	0.086±0.003	0.040±0.005	0.084±0.010
SWT	0.047±**0.003**	0.089±**0.001**	0.026±0.005	0.067±0.005	**0.071**±0.004	0.097±0.029	0.027±0.012	0.071±0.017
SWT+GSFL-WL	0.040±0.005	0.088±0.006	0.026±0.003	0.071±0.010	0.073±0.003	0.078±0.006	0.027±0.008	0.073±0.022
SWT+GSFL-WH	**0.036**±0.004	**0.087**±0.003	**0.024±0.002**	**0.050±0.002**	0.075±**0.002**	**0.075**±0.003	0.018±0.002	0.060±**0.002**
SWT+GSFL-WLH	0.041±0.008	0.092±0.009	0.026±0.007	0.060±0.015	0.076±0.003	0.076±**0.002**	**0.016±0.001**	**0.059**±0.003

TABLE VII
EFFECTS OF DIFFERENT WAVELET BASIS FUNCTIONS. BOLD REPRESENTS THE OPTIMAL OUTCOMES

	Simulated (20 dB)		Samson		Houston		Moffett	
	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
Haar	0.039±**0.001**	0.089±0.003	0.027±0.006	0.056±0.009	0.074±0.003	0.080±0.004	0.022±0.003	**0.058±0.001**
Coif3	0.038±0.003	0.089±0.003	0.025±**0.001**	0.062±0.014	**0.073±0.001**	0.076±0.004	0.017±**0.001**	0.059±0.002
Db3	**0.035**±0.003	**0.086±0.001**	0.029±0.012	0.055±0.010	0.077±0.004	**0.074**±0.005	**0.016**±0.002	0.059±0.003
Sym3	0.038±0.002	0.089±0.002	0.029±0.009	0.058±0.019	0.076±0.003	0.077±0.005	0.019±0.004	0.059±0.003
Bior3.3	0.036±0.004	0.087±0.003	**0.024**±0.002	**0.050±0.002**	0.075±0.002	0.075±**0.003**	0.018±0.002	0.060±0.002

Adding noise to the low- and high-frequency branches at the same time will cause a certain decrease in the precision of unmixing. The reason may be that the low-frequency branch contains less abnormal interference, and the GSFL-W strategy may not play a role or have a counter-effect and interfere with the results. The experimental results show the necessity and design rationality of GSFL-W.

Wavelet basis functions are mathematical functions designed for signal decomposition in WT, and there are many different kinds. Different wavelet basis functions have different expressions, orthogonality, and smoothness. To test the effects of different basis functions on the proposed method, relevant experiments are performed and the results are shown in Table VII. Haar, coif3, db3, sym3, and bior3.3 are commonly used basis functions, where bior3.3 is the basis function adopted by the proposed method. It can be seen from the table that the performance of different basis functions is almost the same in the simulated data, Houston and Moffett. This may be because the design of the wavelet basis function usually considers the generalization of different scenarios, and therefore has relatively little effect on the unmixing results. However, on Samson, bior3.3 performs better than the others. Perhaps, because bior3.3 has biorthogonal characteristics, it can better maintain signal integrity and is more suitable for Samson.

H. Computational Cost

The time cost comparison is listed in Table VIII. The experiments are performed on a desktop computer with a 12th Gen Intel[1] Core[2] i5-1240OF central processing unit, 32 GB of RAM, and an NVIDIA GeForce RTX 4060 graphics processing unit with 8 GB of memory. Compared with other methods, the proposed method does not increase the computational cost significantly and still maintains a fast running speed.

TABLE VIII
COMPARISON OF RUNNING TIME (UNIT: SECOND)

Method	Simulated data (20 dB)	Samson	Houston	Moffett
VCA	0.8	0.7	0.4	0.1
$L_{1/2}$-NMF	10	14	6	1.5
DAEU	53	45	47	40
AAE-Net	49	44	45	38
CyCU-Net	9	8	7	6
MS-Net	16	13	13	11
UST-Net	50	48	52	44
A2SN	7	12	6	5
SWC-Net	12	9	8	6

V. CONCLUSION

In this article, a novel SWC-Net is proposed. The problems of feature-domain extension and effective feature mining are discussed, and the corresponding solutions are given. First, to expand the feature domain, BSPSWT is proposed to divide HSIs into combinations of low-frequency generalized features and high-frequency detailed features while preserving the original 3-D structure, representing different structures of HSIs. Then, to fully exploit the acquired features, SWC-Net is proposed to capture and integrate the spatial–spectral features in different frequency domains through reasonably designed MDFEM, providing more useful information hidden in HSIs. Finally, from the perspective of WT, a generative

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self-supervised feature learning strategy based on wavelet is proposed to better mine effective features, considering that abnormal interference usually exists in high-frequency features. By adding noise perturbation, the network is forced to find more efficient and accurate features to improve the unmixing accuracy. Sufficient comparative experiments demonstrate that the proposed SWC-Net has satisfactory performance. The superiority of WT and the effectiveness of GSFL-W are also proved in the ablation experiments. In the future, our work will focus on further exploring the application of WT in HU.

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